AQ2.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$. If

- $P = -1M_{11} - 2M_{12} + 0M_{13}$
- $Q = -2M_{21} + 0M_{22} + 1M_{23}$
- $R = -1M_{31} - 0M_{32} + 1M_{33}$

where M_{ij} is the minor with respect to the (i, j)-th entry, then choose the set of correct options.

- [ ] $det(A) = -2$.
- [ ] $P \neq Q$.
- [ ] $P \neq Q \neq R$.
- [ ] $P = Q = R$.
- [ ] $Q \neq R$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -2$.
$P = Q = R$.

1 point

Suppose for a real 3×3 matrix A, there exists a real 3×3 matrix P such that $D = PAP^{-1}$ is a real 3×3 diagonal matrix. Choose the correct set of options.

- [ ] $det(A)$ must be equal to $det(P)$.
- [ ] $det(A)$ must be equal to $det(D)$.
- [ ] The matrix A must be equal to D.
- [ ] If D is the identity matrix of order 3, then A must be the identity matrix of order 3

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A)$ must be equal to $det(D)$.
If D is the identity matrix of order 3, then A must be the identity matrix of order 3

1 point

Choose the set of correct options for square matrices of order 3.

- [ ] If AB = 0AB = 0, then one of the matrices between AA and BB must have determinant 00.
- [ ] If AB = 0AB = 0, then both the matrices AA and BB must have determinant 00.
- [ ] If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $det\,A$ = $\frac{3}{det(B)}$ detA = det(B)3.
- [ ] If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $det\,A$ = $\frac{9}{det(B)}$ detA = det(B)9.
- [ ] If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $det\,A$ = $\frac{27}{det(B)}$ detA = det(B)27.
- [ ] If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then both AA and BB must have non-zero determinant.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If AB = 0AB = 0, then one of the matrices between AA and BB must have determinant 00.
If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $det\,A$ = $\frac{27}{det(B)}$ detA = det(B)27.
If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then both AA and BB must have non-zero determinant.

1 point

Consider a matrix A = $\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$abcbcacab. If $a + b + c$a + b + c is divisible by 66 then choose the set of correct options.

- [ ] $det(A)$det(A) is divisible by 55.
- [ ] $det(A)$det(A) is divisible by 66.
- [ ] $det(A)$det(A) is divisible by 22.
- [ ] $det(A)$det(A) is divisible by 33.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A)$det(A) is divisible by 66.
$det(A)$det(A) is divisible by 22.
$det(A)$det(A) is divisible by 33.

Let A = [a_{ij}]A = [aij] be a square matrix of order 3, where a_{ij} = $i$aij = i. Find $det(A)$det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 0
1 point

Level 2:

Let $A = [\delta_{ij}]$A = [δij] be a square matrix of order 33, where $\delta_{ij} = i \times j$δij = i × j. Find $det(A)$ det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
1 point
Consider the following square matrices:

$A = \begin{bmatrix} 201320142015 \\ 201620172022 \\ 201920202021 \end{bmatrix}$, $B = \begin{bmatrix} 201620172022 \\ 201320142015 \\ 201920202021 \end{bmatrix}$ and $C = \begin{bmatrix} 403240344044 \\ 201320142015 \\ 201920202021 \end{bmatrix}$A = $\begin{bmatrix} 201320162019 \\ 201420172020201520222021 \end{bmatrix}$, B = $\begin{bmatrix} 201620132019201720142020202220152021 \end{bmatrix}$ and C = $\begin{bmatrix} 403220132019403420142020404420152021 \end{bmatrix}$

Choose the set of correct options.

- [ ] $det(B) = det(A)$det(B) = det(A).
- [ ] $det(A) = -det(B)$det(A) = −det(B).
- [ ] $det(C) \neq -2det(B)$det(C) = −2det(B).
- [ ] $det(C) = -2det(A)$det(C) = −2det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -det(B)$det(A) = −det(B).
$det(C) \neq -2det(B)$det(C) = −2det(B).
$det(C) = -2det(A)$det(C) = −2det(A).

(Based on the below information answer questions 8 and 9).

Let AA be a real 3 × 33 × 3 matrix as given below:

$A = [C_1\ C_2\ C_3]$A = [C1C2C3]

where C_i Ci represents the ii-th column of the matrix AA. Now let us consider the following set of matrices.

- A_1 = [C_1 C_2 + 5C_3 C_3]A1 = [C1C2 + 5C3C3]
- A_2 = [C_1 + C_2 + C_3 C_2 C_3]A2 = [C1 + C2 + C3C2C3]
- A_3 = [C_1 C_2 + 5C_3 0]A3 = [C1C2 + 5C30]
- A_4 = [C_1 + C_2 C_2 + C_3 C_3 + C_1]A4 = [C1 + C2C2 + C3C3 + C1]

1 point

Which of the matrices have the same determinant as that of AA?

- [ ] A_1 A1
- [ ] A_2 A2
- [ ] A_3 A3
- [ ] A_4 A4

No, the answer is incorrect.

Score: 0

Accepted Answers:

A_1 A1

A_2 A2

1 point

Choose the set of correct options.

- [ ] $det(A_3)$det(A3) cannot be determined from the given information.
- [ ] $det(A_3)$ = 0det(A3) = 0
- [ ] $det(A_4)$ = 2 $det(A)$det(A4) = 2 det(A)
- [ ] $det(A_2)$ = 3 $det(A)$det(A2) = 3 det(A)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A_3)$ = 0det(A3) = 0

$det(A_4)$ = 2 $det(A)$det(A4) = 2 det(A)

If the diagonal entries of a 3 × 33 × 3 lower triangular matrix AA are 1,2, and 3, then find the sum of roots of the equation $det(A - xI)$ = 0det(A − xI) = 0, where II is the identity matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Check Answers

Your score is: 0/10